2017 H2 Chemistry 9729 Preliminary Examinations
Suggested solutions

Paper 2: Structured Questions

1 (a) To drive off the CO_2

(b) (i) Amount of HCl = 50.0 x 10^{-3} x 0.100 = 5.00 x 10^{-3} mol

(ii) Amount of HCl = 20.0 x 10^{-3} x 0.0500 = 1.00 x 10^{-3} mol

(iii) Amount of HCl unreacted in graduated flask = 2 x 1.00 x 10^{-3} = 2.00 x 10^{-3}mol
Amount of HCl neutralised = 5.00 x 10^{-3} – 2.00 x 10^{-3} = 3.00 x 10^{-3} mol

(iv) Amount of excess stomach acid
= (375 x 10^{-3} x 0.0300) – (375 x 10^{-3} x 0.000300) = 0.011138 mol

1 tablet is able to neutralise 3.00 x 10^{-3} mol of HCl.
To neutralise 0.011138 mol of HCl, minimum of 4 tablets is needed.

(c) Amount of $CaCO_3$ in 1 tablet = ½ x 3.00 x 10^{-3} = 1.50 x 10^{-3} mol

Mass of $CaCO_3$ in 1 tablet = 1.50 x 10^{-3} x 100.1 = 0.150 g = 150 mg.

There are only 150 mg of $CaCO_3$ in 1 tablet.

(d) **(i)** For Al_2O_3, there is a **decrease** in **ionic character/ increase in covalent character**

due to the **high polarising power of Al^{3+} ion** which distorts the electron cloud of O^{2-} ion to a greater extent compared to Mg^{2+} ion. Thus the melting point is lower than MgO.

(ii) MgO is only **slightly soluble in water.** pH = 9 (accepts 8 to 10)

Al_2O_3 remains **insoluble** in water. pH = 7.

(iii) $Al^{3+}(aq) + 3OH^-(aq) \rightarrow Al(OH)_3(s)$
$Al(OH)_3(s) + OH^-(aq) \rightarrow Al(OH)_4^-(aq)$

2 **(a)** The volatility of the halogens **decreases** down the group.

The halogens have simple molecular structure and are non-polar.

The **instantaneous dipole-induced dipole interactions between the molecules of each halogen become stronger** due to the **increasing size and polarisability of the electron cloud of the halogens down the group.**

Hence more energy is required to overcome the stronger instantaneous dipole-induced dipole interactions between the molecules of each halogen.

(b)

hydrogen halides	H−C*l*	H−Br	H−I
bond energy / kJ mol^{-1}	431	366	299

The thermal stability of the hydrides of the Group 17 elements **decreases** down the group as the **H−X bond strength decreases** down the group. Hence less energy is required to overcome the H−X covalent bond.

(c) **(i)** $K_{sp} = [Pb^{2+}][Cl^-]^2$ mol^3 dm^{-9}

correct expression and units

(ii) $[PbCl_2] = 4.7 \div 278.2 =$ **0.016894 mol dm^{-3}**

$K_{sp} = (0.016894) \times (2 \times 0.016894)^2 =$ **1.9×10^{-5}** mol^3 dm^{-9}

[1] for correct calculation of solubility in mol dm^{-3}
[1] for correct numerical answer and leave the answer to 2 significant figures

(d) **(i)** Decolourisation of orange/yellow/reddish brown Br_2(aq) and a black solid is formed.

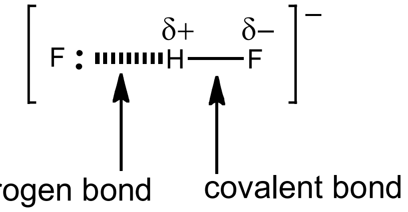
$Br_2(aq) + 2At^-(aq) \rightarrow 2Br^-(aq) + At_2(s)$

(ii) no reaction

(e) **(i)** cation: K^+

anion: HF_2^-

(ii)



[1] for covalent bond; [1] for hydrogen bond

[Total:12]

3 **(a)** **(i)**

- Mg^{2+} has a smaller ionic radii and thus a higher charge density and polarising power.
- Ease of distortion of electron cloud of carbonate increases, weakening of C–O covalent bond occurs to a greater extent.
- thermal stability of magnesium carbonate is lower than calcium carbonate.

(ii) $MgCO_3 \rightarrow MgO + CO_2$

(b) $Mass\ \ of\ \ CO_2 = 1.00 - 0.777 = 0.223g$

$$n_{CO_2} = \frac{0.223}{(12 + 16 \times 2)} = 5.0682 \times 10^{-3} mol$$

$$n_{ZCO_3} = 5.0682 \times 10^{-3} mol$$

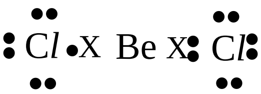
$$5.0682 \times 10^{-3} = \frac{1.0}{M_{MCO3}} \quad \Rightarrow \quad M_{ZCO_3} = 197.3 gmol^{-1}$$

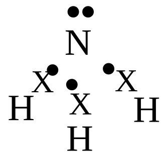
$$M_Z + (12 + 16 \times 3) = 197.3$$

$$M_Z = 137.3 gmol^{-1}$$

Identity of **Z** : **Barium**

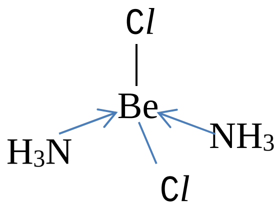
(c) (i)





Both correct [1]

(ii)



Tetrahedral shape around Be.

[Total:10]

4 (a) (i) Amount of coal required

$= \dfrac{108 \text{x} 10^3}{393}$

$= 274.81 \text{mol}$

Mass of coal required

$= 274.81 \text{x} 12.0$

$= 3297.7 \text{g}$

Energy density of coal

$= \dfrac{108}{3297.7}$

$= 0.0327$ MJ/g

(ii)
- Coal is a **more efficient fuel.**
- The energy density of coal (0.0327 MJ/g) **is larger than the energy of density of ethanol.** Thus for the **same amount of energy supplied**, **lesser amount of coal is required.**

(iii)
- CO undergoes **<u>ligand exchange reaction</u>** and **<u>binded more strongly via dative bond</u> to haemoglobin than O_2**.
- Thus, this **<u>displaces O_2 from oxyhaemoglobin</u>** and **<u>reduces the amount of haemoglobin available for carrying O_2.</u>**

(b) (i) $2C\,(s) + 3H_2\,(g) + \frac{1}{2}\,O_2\,(g) \rightarrow CH_3CH_2OH\,(l)$

$\Delta H^{\ominus}_f$ □ =□$\Delta H^{\ominus}_c$ (C) + 3□$\Delta H^{\ominus}_c$ (H_2) + ½ $\Delta H^{\ominus}_c$ (O_2) – $\Delta H^{\ominus}_c$ (CH_3CH_2OH)

= 2 x (–393) + 3(–286) + 0 – (–1367)

= **–277 kJmol^{-1}** [1]

(ii) $\Delta H^{\ominus}_r$= Energy absorbed to break bonds + Energy released to form bonds

CH_3CH_2OH (*l*) + 3 O_2 (g) → 2 CO_2 (g) + 3 H_2O (*l*)

bonds broken	(endothermic)	bonds formed	(exothermic)
5 × C–H	**5 × (+410)**	**4 × C=O**	**4 × (−805)**
1 × C–O	**1 × (+360)**	**6 × O–H**	**6 x (<u>−460</u>)**
1 × C–C	**1 x (+350)**		
1 × O–H	**1 x (+460)**		
3 O=O	**3 × (+496)**		
total energy absorbed	**+4708**	**total energy evolved**	**− 5980**

$\Delta H^{\ominus}_c$ = +4708 – 5980
$\Delta H^{\ominus}_c$ = **– 1272 kJ mol^{-1}**

- ✓ Bond energy data from data booklet [1]
- ✓ Final answer + units [1]

(iii) ✓ **Bond energy values in the Data Booklet are averages**, so it would not apply exactly to a particular compound.

Or

✓ In the calculation in **(b)(ii)**, it is assumed that all the reactants and products are in the gaseous state. However, **ethanol is a liquid at standard conditions**

(c) (i)
Anode: $CH_3CH_2OH + 3H_2O \rightarrow 2CO_2 + 12H^+ + 12e^-$
Cathode: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$
Overall: $CH_3CH_2OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$

(ii) $E^{\ominus}_{cell} = E^{\ominus}_{cathode} - E^{\ominus}_{anode}$
= +**<u>1.23</u>** – (+0.08)
= **+ 1.15V**

✓ Correct value for $E^{\ominus}_{cathode}$ [1]
(allow for e.c.f if student wrote the wrong equation at the anode as :
$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$, $E^{\ominus}_{cathode}$ = +0.40)
✓ Correct value for $E^{\ominus}_{cell}$ [1]

(iii) $\Delta G^{\ominus} = -nFE^{\ominus}_{cell}$
= –12 x 96500 x 1.15
= –1.33 x 10^6 Jmol^{-1}

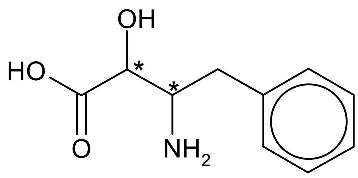
= –1.33 x 10^3 kJmol^{-1} (correct units)

(d) (i) **H_2O(g) (steam), H_3PO_4 catalyst, heat, high pressure**

(ii)
1. In ethene, each of the carbon atom is **sp^2 hybridised**.
2. Each carbon atoms forms **two σ-bonds with hydrogen atoms** through **the head on overlap** of two of **its sp^2 orbitals with the 1s orbitals** of the **hydrogen atoms**.
3. Another **σ-bond** is formed when the remaining **sp^2 orbital of the two carbon atoms overlap with each other**.
4. The **remaining unhybridised p orbital** of the two carbon atoms, which is perpendicular to the plane of sp^2 hybrid orbitals, **overlaps laterally or sideways to form a π bond.**

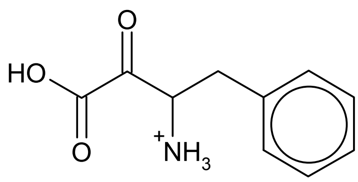
5 (a) (i) carboxylic acid , (secondary) alcohol , and (primary) amine .

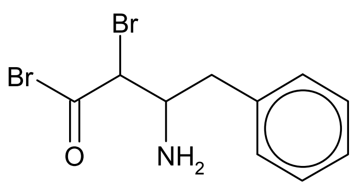
(ii)



total possible number of stereoisomers = 2^2 = **<u>4</u>**

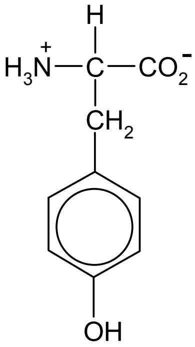
(iii)





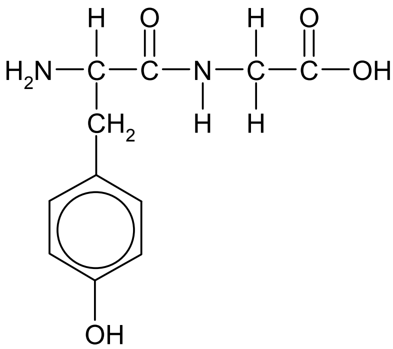
(b) **(i)** A zwitterion is a molecule that is **<u>neutral overall</u>**, but **<u>contains a positive charge</u>** on one part of the molecule (usually $-NH_3^+$) **<u>and a negative charge</u>** on another part of the molecule (usually $-CO_2^-$).

(ii)



(iii) peptide or amide linkage

(iv)



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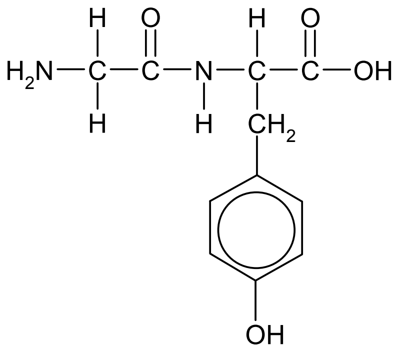
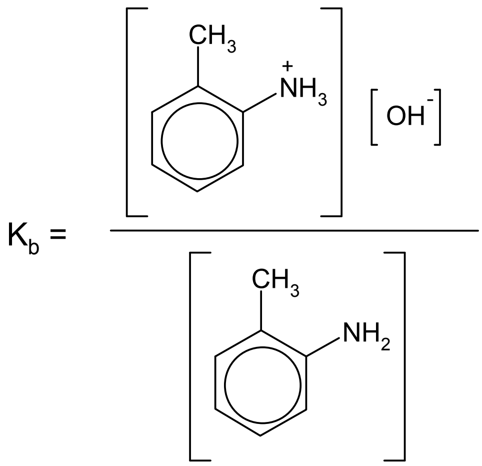


(c) **(i)** NaOH(aq), reflux
or
HC*l*(aq), reflux, followed by NaOH(aq)

(ii)



(iii) $K_b = 10^{-pK_b} = 10^{-9.6} = 2.5119\times10^{-10}\ \text{mol dm}^{-3}$

$[OH^-] = \sqrt{K_b \times c_{base}} = \sqrt{2.5119\times10^{-10}\times0.10} = 5.0119\times10^{-6}\ \text{mol dm}^{-3}$

$pOH = -\log[OH^-] = -\log(5.0119\times10^{-6}) = 5.30$

$pH = 14.0 - 5.30 = 8.70$

(d) The pK_b of benzylamine will be **lower** than that for 2-methylphenylamine

Benzylamine is a stronger base as **the phenylmethyl group is electron donating**, **making the lone pair of electrons on N more available for donation** to a proton (or H^+ ion).

2-methylphenylamine is a weaker base as the **lone pair of electrons on the N** of the amine is **delocalised into the benzene ring**, making them **less available for donation** to a proton (or H^+ ion).